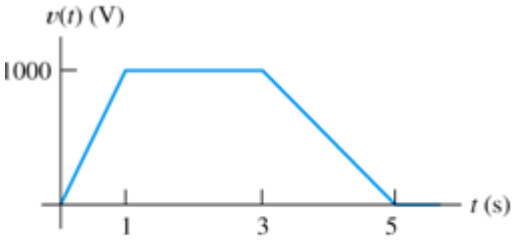
Chapter 8 Tutorial Exercise

Question 1

Suppose that the voltage waveform shown in Figure (a) is applied to a 10-μF capacitance. Find and plot the current, the power delivered, and the energy stored for time between 0s and 5s.



(a)

Question 2

(a) Two capacitance of 2-μF and 1-μF are in series. Find the equivalent capacitance.
(b) Repeat if the capacitances are in parallel.

Question 3

We want to design a 1-μF capacitor. Compute the length required for rectangular plates of 2cm width if the dielectric is polyester of 15-μm thickness.

Question 4

The current through a 10-mH inductance is $i(t) = 0.1\cos(10^4 t)\ A$

Find the voltage and stored energy as functions of time.